When does a river forecast change a decision?

Flood warning at Gandak Barrage and water allocation on the Potomac

Abstract

River forecasts can affect the timing of flood warnings and the amount of water promised to users. We evaluated these decisions in two historical case studies. At Gandak Barrage, India, we compared forecasts using recent inflow, an upstream gauge at Devghat in Nepal, and archived rainfall forecasts. On 25 September 2026, the gauge and rainfall together signalled a study-defined high-flow threshold six hours before it was crossed. The gauge alone gave four hours of warning. Only one qualifying onset occurred. For the subsequent crest, the combined model predicted 22,780 m³/s against an observed 15,396 m³/s. At 24 hours, carrying the last inflow forward also gave a smaller mean absolute error than the combined model. On the Potomac River, we simulated daily allocations using six years of forecasts. Planning for a low flow reduced the frequency of overcommitment from 63.5% to 1.5%, at the expense of greater unmet demand. Under a linear cost assumption, that choice was preferable when the cost per unit of overcommitment exceeded 3.55 times the cost per unit of unmet demand. These results distinguish three outcomes a forecast user might value differently: earlier warning, an accurate crest estimate, and a reliable water commitment.